# FINITENESS GOES UP IN TOWERS OF ALGEBRAIC EXTENSIONS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: The dimension-monotonicity half of tower bookkeeping, complementing the multiplicativity-of-degree lemma (delivered L48338).

*Editorial note.* The label lemma-finite-goes-up names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Definition of a finite extension: finite-dimensional as a vector space over the base
- For a tower F inside E inside K, the E-linear span of a set contains its F-linear span

**Lemma 0.1.** *Let $K/E/F$ be a tower of algebraic field extensions. If $K$ is finite over $F$, then $K$ is finite over $E$.*

*Proof.* Direct from the definition. □

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